

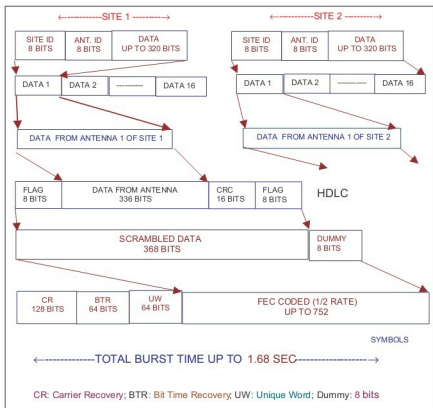


Fig. 5: Transmission burst format (channel frame structure)

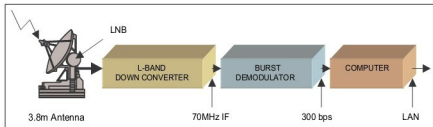


Fig. 6: Receive ground hub station

ent monitoring points of a landslide-prone site using multiple antennae (see Fig. 3) and a single GPS receiver. This real-time data collected at different landslide-prone sites, situated at different parts of the entire country, can be transferred to a remote central monitoring centre using a suitable satellite-based communication network system (see Fig. 2).

GPS with multiple antennae

Usually, several observation points must be monitored to fully understand the stability and any ongoing deformation that could cause slope

failure. For example, it may require 12 to 16 observation points to adequately monitor a high-risk slope measuring 300×500 m².

A land-survey-type GPS receiver offers greater accuracy and is highly automated and less labour-intensive than conventional techniques used during the stability monitoring of high-risk slopes.

Fig. 3 shows a GPS multi-antenna system. It uses an electronic switching device for multiple-antenna GPS deformation-monitoring system. It sequentially allocates time to each antenna. The main parameter entered into this unit is the time allotted to an antenna

in each round of measurements. The receiver makes slandered pseudo-range and carrier-phase observations for each antenna.

When the rate of deformation of the monitored feature is low, the integer ambiguities for each epoch may be determined based on the previously known coordinates of the monitored points. Alternatively, for high-accuracy measurements the receiver could remain connected to each antenna for a set period of time (10 minutes with receiver-sampling interval of 15 seconds) to allow it to acquire a number of epochs of data from an antenna before it switches to the next. The data from each antenna can be processed as a short-time series using standard double-differencing techniques with initial-ambiguity resolution. Processed data output of each antenna, every 10 minutes, would be 320 bits, i.e., 40 characters (8 bits/character).

If the required distance between the antenna and the unit is up to 30 metres, coaxial cables may be used and for more than 30 metres, a low-noise amplifier or fibre-optic cable can be used. It should have the facility to change switching for a number of antennae (5 to 16) using the keypad mounted on it.

Each site should have a reference base station which sends data to the hub station through a satellite using separate transmit unit (see Fig. 3).

Satellite-based communication network

For relaying GPS data from different sites (at different parts of country) to a central monitoring hub station, the proposed satellite-based communication system is shown in Figs 2 and 8.

The real-time data collected by the multi-antenna GPS system at different sites is transferred to the satellite transmit unit (UHF transmitter). The satellite transmit unit (UHF transmitter) receives data from the GPS receiver. Then this data is proc-